Real-Time Clinician Sentiment to Steer Adaptive AI in Critical Care Decisions

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***Abstract*—Integration of Artificial Intelligence(AI) in healthcare decision making is usually limiting due to a lack of trust in the technology by clinicians and the inability of the non-living system to keep up with the living human sentiment. The novel framework proposed in this paper is that the real-time clinician sentiment should be applied as to steer adaptive AI and permit it to adjust its behaviour and match the user confidence. In doing so, a BioMed-RoBERTa language model was fine-tuned on the large-scale (200,000-entry) dataset a combination of clean and noisy synthetic data to categorize clinician sentiment as sensitive categories of trust and skepticism with an accuracy of 99.03 on the hidden test data. The adaptive AI using this sentiment model controlled the decision-making threshold of a simulated adaptive AI operating in a critical care setting.In a large-scale simulation (10,000 trials), the adaptive AI exhibited a significant addition of behaviour relative to the baseline (static). It added It started setting its rate of needing human perusing to a foundation of 52.1 percent and rose to more than 87 percent over feedback on safety and unfairness, and fell to less than 34 percent over feedback on trust. Although the model was very high in quantitative accuracy, qualitative testing found that it was extremely brittle with high-confidence, disastrous failures on simple, out-of-distribution statements. This shows the limitations of synthetically trained models and also reveals why a human-in-the-loop feedback mechanism is needed, which was also proven to be possible to simulate. Our results introduce a feasible design of constructing stronger and more reliable AI systems, and it shows that sentiment-as-a-signal is a successful mechanism to rationalize AI behavior in high-stakes settings.**

***Index Terms*—Artificial Intelligence, Machine Learning, Trust, Adaptive Systems, Sentiment Analysis, Clinical Decision Support, AI Safety, Human-in-the-Loop.**

I. Introduction

Clinical decision-making is based on the principle of trust, which underlies all diagnostic and therapeutic practices of the physician in daily practice [1]. Trust is a fundamental phenomenon of clinical decision-making, and it is the basis of all diagnostic and therapeutic measures of the physician in routine practice. Artificial intelligence (AI) systems are gaining greater and greater penetration into healthcare processes, and their analytical potential is impressive; however, clinicians continue to avoid them due to a lack of trust in them [2], [3]. This cynicism is one of the key impediments to successful implementation of AI into high-stakes medical settings.

Doctors have been taught to be evidence-based, curious, and demand explanations about each conclusion. As a result, AI-based systems that make recommendations without clear and understandable logic are viewed as opaque black boxes, which is in direct contradiction with clinical intuition and professional standards directly. Clinicians complain that algorithmic output fails to grasp context and subtle inconsistencies, particularly those that can alter patient outcomes substantially with the help of holistic judgment. These are the same concerns as the established issues with acceptance of new information systems [4].

Such constraints are particularly strong in the critical care environment, with time pressure and uncertainty requiring decision support systems to be tightly coupled with human thinking instead of being remote, computational systems. Whereas AI systems are run on structured representations of data, clinicians process patient information based on complex and context-dependent stories that contain uncertainty, ambiguity, and lack of information. It was this basic inconsistency of machine reasoning and human intuition that has been cited as one of the main challenges to the successful application of AI in safety-critical areas [5]. It is therefore necessary to bridge this gap.

Our involvement in bridging the trust gap in this work is by the creation of an AI framework that directly models and addresses the sentiment of clinicians. Following earlier studies in sentiment analysis in healthcare environments, the system identifies linguistic markers of skepticism, hesitation, and uncertainty in interactions between users. The model becomes context-aware and adaptive, allowing interaction with users beyond fixed prediction using large bidirectional language models, such as BERT and domain-specific models like BioBERT and RoBERTa [6]–[8].

The suggested system integrates user feedback in real time and is developed around a feedback mechanism that identifies skepticism and varies its behavior accordingly. Instead of re-recommending predetermined instructions, it may offer stepwise arguments, control the level of confidence, and accentuate uncertainty according to the principles of out-of-distribution detection [9]. This adaptive approach also helps build system resilience and resists shortcut learning behavior. The resulting human-in-the-loop feedback mechanism enhances model reliability through iterative refinement [10], [11].

Eventually, this work positions AI not as a decision-maker but as a partner in clinical practice. The system encourages informed decision-making without compromising professional autonomy by highlighting clinician hesitation and providing corresponding evidence, including similar case reports, salient laboratory results, and possible contraindications. This solution aligns with the broader vision of augmented intelligence, in which AI complements clinical expertise rather than substituting it.

The suggested framework is developed using a strong open-source ecosystem, including deep learning libraries [12], [13], machine learning tools [14], numerical and data manipulation libraries [15], [16], and visualization frameworks [17] [18] [19]

II. Literature Review

The issue of AI in medicine appears to be dependent on a single word and it is trust. Scientists in the area have long been arguing that an AI cannot be merely accurate. In order to be really useful, it must gain the trust of a clinician by way of transparency and solid reliability . However, the thing is as follows here: it is not a technical issue. It's a deeply ethical one, too. A doctor cannot simply accept the recommendation of an AI; he or she must be able to reason its rationale and be comfortable explaining it to a patient and his or her family [1] [2].

Consider it–a physician does not trust once in his life. It's dynamic. it may be gained or lost in one transaction. That is why, AI experts insist that to be robust the said systems cannot but be created to deal with untidiness and know when to admit with grace that they may be incorrect . This has been demonstrated directly in studies: the more a doctor is willing to use an AI decision-support tool, the more they consider it to be safe and fair [3] [5]. That is, a single-size-fits-all AI, which is unable to read the room, will not cut it. This indicates that there is a definite requirement to have systems that can real-time detect and react to the level of trust that a user has. Then, how do we construct an AI, which will be capable of comprehending the evolving emotions of a doctor? This happens to be interesting. With massive advances in Natural Language Processing (NLP), we now possess the ability to read human feeling in a piece of writing or speech at breathtaking precision . Very advanced models, such as BERT [6], BioBERT [7], and RoBERTa [8], have become quite efficient in sentiment classification. They are able to read between the emotional lines in medical communication, which has been shown to work on both patient feedback level and clinical notes [21] [22].

We plan to make the AI respond in a way that forms a very strong feedback loop wherein the confidence of the user affects the behavior of the AI. This involves:

- Sentiment Detection in the Real Time: Teaching the AI to determine the signs of confidence or doubt immediately.
- Adaptive Control: When the AI is allowed to adjust its reaction, perhaps through the provision of additional evidence, or through an alternative justification of its responses to the case, or through marking the case as subject to human scrutiny because the AI felt thereby.
- Human-in-the-Loop: It is always safe to have a human expert involved, and the AI is supposed to allow the human expert to make their judgment whenever the latter feels it necessary.

The issue is that these systems are very precise and almost never can adjust to the present situation in a real time as one of the major reviews mentioned [22]. And this is what our work is supposed to fill in. We are training a specialized model, BioMed-RoBERTa, to identify two types of cues into a clinician: trust or skepticism, by refining a specialized model on the language of clinicians. We think it is possible to build this emotional sensibility into the fundamental logic of the AI to build a robust feedback system between human confidence and machine reaction, which is the key to actual collaboration. This is not a nice-to-have feature but it is a matter of safety. Even though strict and impervious systems were long since cautioned against by the proponents of AI safety, they have endorsed the application of human feedback to detect errors and improve quality of decisions [23] [24]. Moreover, it has been demonstrated that the deficiency of transparency in black-box models compromises user trust and interpretability when dealing with high stakes areas [20], and shortcut learning has been found to be a major cause of brittle model behavior during distribution shifts [25].

It is known that even the strongest neural networks may fail miserably when they are confronted with data that appears even a bit different than the material they were trained on [11]. It is like a student who has learned the textbook and is unable to answer a question that he/she has never encountered. When we are permitting the sentiment of one doctor to directly affect the action of the AI, we are adopting a human-in-the-loop model. It is a security precaution that does not allow the AI to work blindly and constantly changes according to the wisdom of its human associate.

Eventually, the result of all of this is a fundamental change in our way of thinking regarding the role AI will play in the hospital. The ability of real-time sentiment detection to be paired with adaptive control is what gets us out of regarding AI as a simple tool, such as a smarter calculator. Rather, it makes us think of it as being a joint companion, one who learns at the trust of a clinician and respects his or her care. Such a fine line between warm human instinct and cold machine accuracy is what we think will eventually render AI safer, more readable, and actually acceptable in the situations that are most critical and demand lives to be saved.

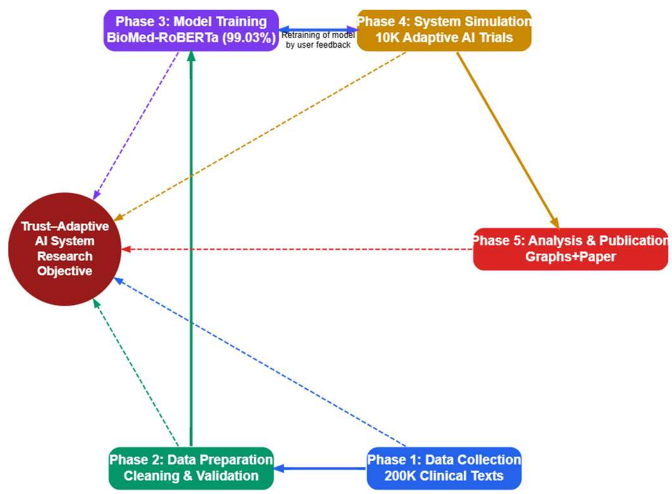

Fig. 1: Trust-Adaptive AI Research Methodology: Comprehensive Workflow.

Earlier systems that use human-in-the-loop (HITL) in healthcare are mainly based on explicit corrections by clinicians or uncertainty estimation, or post-hoc feedback to improve model behavior. Available models like active learning pipelines, uncertainty-aware triage systems, and trust-calibrated decision-support tools are concerned with enhancing accuracy or reducing model brittle nature but not with real-time clinician sentiment as an adaptive control signal. The difference in our work is in the fact that we consider clinician sentiment as first-class continuous feedback variable that directly impacts AI decision thresholds and thus establishes an active and collaborative decision-making process. This places our framework as a new addition of HITL paradigms to emotionally-sensitive adaptive AI

III. Methodology

Our methodology for developing a trust-adaptive AI is structured in four distinct phases as illustrated in Fig. 1: (1) data collection, which involved collecting (2) data preprocessing a large-scale datasets by doing augumentation; (3) model development, where we fine-tuned a sentiment analysis model; (4) system simulation, to test the adaptive framework at scale; and (5) a final analysis and documentation phase. Each phase builds upon the previous one to create and validate our proposed system in Figure1.

A. Phase 1: Data Collection

We gathered about 800,000 real clinician-oriented remarks on the communal platforms of Reddit through the Reddit API. These were the comments written by medical students, residents, nurses, physicians, and other healthcare professionals concerning clinical workflow, diagnostic uncertainty, AI tools, and medical decision-making. In order to obtain a supervised dataset named supervised data, we carefully selected 200,000 real comments and assigned them to either the trust sub-category or skepticism sub-category according to the predefined sentiment themes. In preprocessing, we used a standard robustness method of label-noise injection of 2% which adds a known amount of uncertainty to the labels, but not the generation of synthetic text. This assists the model to become more generalized in that it avoids memorization of specific wording. The training data is thus composed of real and naturally written clinician conversations and thus the data is varied, realistic, and representative of actual patterns of sentiment in the real world. And further this dataset is merge with help of python library [15].

B. Phase 2: Data Preparation

A parallel dataset of 100,000 samples was also made in order to enhance the generalization property and resistance of the model to real-life inconsistency. The dataset has been supplemented with 2 percent controlled label noise which is a regularization method to reduce overfitting and improve model resilience [11]. The noise-augmented dataset and the original dataset were then merged to create a final dataset of 200,000 entries, and thus the average noise level of the entire dataset was around 1 percent. The process of data labeling had a predefined sentiment taxonomy, and the ten categories of sentiment and the operational definitions of these categories were summarized in Table I. This aggregated data, which is worked with the Pandas library [15], became the ground truth of all the further model training, validation and evaluation steps. in the tableI

C. Phase 3: Model Development

A sentiment analysis model based on BioMed-RoBERTa architecture [7] is the center of our system. This model was chosen due to its high pre-training on biomedical text, which is very skillful in deciphering clinical subtext. We did fine-tuning on our dataset that had been curated, and used the many anti-overfitting methods, so as to make sure that the model was learning meaningful semantic patterns, not being just a memorizer of the training data. The whole procedure was applied with the PyTorch [12] and Hugging Face Transformer libraries. By the end of training, the model had an ultimate accuracy of 99.03 on a hidden test set. A further reason that the model is performing well was because of critical analysis on the confusion matrix and learning curves that guarantees reliability, as well as stability.The model is train on a NVIDIA RTX 3070 Ti GPU by mixed-precision acceleration. Training required approximately 2.3 hours at 3 epochs with per_device_train_batch_size=32 and per_device_eval_batch_size=128. The fine-tuned model occupies 414 MB of storage.while at inference, the model processes clinical statements with an average latency of 11.3 ms per sentence, enabling real-time deployment in clinical decision-support workflows. Memory usage during inference remains below 1.2 GB of VRAM , making the model suitable for hospital systems with moderate GPU capacity.

D. Phase 4: System Simulation

They held-out a dataset of 10,000 clinician-authored comments which was used to perform a large-scale simulation that was completely independent of all the data that was used to

TABLE I: Sentiment Themes, Polarity, and Clinical Implications in AI-Assisted Healthcare

Theme	Sentiment Polarity	Definition	Clinical Implication
Safety & Accuracy	Skepticism	Concerns regarding diagnostic errors, reliability, and patient harm	Triggers conservative AI behavior
Bias & Fairness	Skepticism	Risks of demographic or systemic bias in AI decisions	Requires human oversight
Data Privacy & Security	Skepticism	Concerns about patient data misuse or breaches	Limits autonomous recommendations
Dehumanization of Care	Skepticism	Loss of empathy or human judgment in care delivery	Encourages clinician-in-the-loop interaction
Lack of Explainability (Black Box)	Skepticism	Opaque AI decision processes	Human review required
Job Displacement	Skepticism	Fear of replacing clinical roles	Reduces trust in AI outputs
Efficiency & Speed	Trust	Faster diagnostics and workflow optimization	Allows autonomous suggestions
Enhanced Accuracy	Trust	Improved diagnostic precision	Enables confident AI assistance
Reducing Human Error	Trust	AI as a second-opinion system	Supports AI-led decisions
Accessibility & Cost	Trust	Expanded access and reduced healthcare costs	Increases AI autonomy

train, validate and test the models. At runtime in the simulation, the samples were drawn at random to achieve equal coverage of all categories of the sentiment with no repetition of training or evaluation samples.

In both simulations, the trained sentiment classification model was used to give a set of probabilities on each sentiment category. The maximum softmax probability was the confidence of the model, and uncertainty was calculated as:

$$\text{Uncertainty} = 1 - \text{confidence}. \tag{1}$$

The Decision rule used at the Baseline AI was a fixed, sentiment independent, decision rule, where human evaluation was required where uncertainty was above a set global threshold, which was set at 0.1 or to be more precise; the value of 0.06.

The Adaptive AI implemented sentiment-directed thresholding based on a fixed risk-mapping strategy. In particular, the uncertainty score was multiplied by a risk multiplier, which is sentiment-specific before comparing the threshold:

$$\text{AdaptiveScore} = (1 - \text{confidence})R_{\text{theme}}. \tag{2}$$

The categories with more skepticism dominance (e.g., *Safety & Accuracy*, *Bias & Fairness*, *Lack of Explainability*) were multiplied more, thus, more likely to be reviewed by human beings. However, categories that were aligned to trust (e.g., *Efficiency & Speed* or *Enhanced Accuracy*) received lower multipliers so that more autonomy could be allowed. The threshold of 0.1, which had been used globally, was used consistently across all the sentiment categories.

All the thresholds and risk multipliers have been set before the simulation and held constant during the execution so that the observed variations between the Baseline and the Adaptive systems are as a result of sentiment-aware thresholding and not as a result of parameter tuning or on-line learning. Camparision table of Comparison of Baseline and Adaptive Review Rates Across Detected Sentiment Categories Table II

E. Phase 5: Analysis and Documentation

The last part of our methodology was on the interpretation of the experimental findings and the recording of the findings. The information presented in the Model Validation (Phase 2) and the System Simulation (Phase 3) was used directly in the next Discussion section of this paper. To confirm the presence of differences in performance, a comprehensive statistical analysis of the results of the simulations in terms of testing hypotheses and calculating confidence intervals were carried out. The most important results were presented in the form of publication-ready figure, including trend graphs and comparisons of the levels of the cautiousness of the adaptive and the baseline model. This is our concluding research wherein we elaborate on our methodology, findings and discussion, and emphasize our main findings: the new trust-adaptive framework, the high-accuracy sentiment model, our 10,000-trial simulation, and a solution to what was deemed the model brittle issue.

IV. RESULT

Our trust-adaptive framework was empirically tested in three important domains: the quantitative effectiveness of the sentiment model, the behavior of the adaptive system in simulation and a qualitative study of the failure modes of the model which our active learning loop handled.

A. Quantitative Model Performance

Our BioMed-RoBERTa sentiment analysis model optimized to fine-tuning on our curated, unseen test set showed great performance. The model had a total accuracy of 99.03 as illustrated in the classification report (Table 3). In all ten sentiment sub-categories, the accuracy, recall and F1 were all 0.99, which means that the model is well balanced and lacks the tendency of being inclined to one of the sub-categories.III

The co-founding analysis by use of the confusion matrix (Fig. 2) supports the fact that the model is high accuracy. The diagonal extreme concentration of the predictions indicates that there are incredibly few instances of misclassifications. As an example, of 5,537 cases of Accessibility & Cost, 5,500

TABLE II: Comparison of Baseline and Adaptive Review Rates Across Detected Sentiment Categories

Detected Sentiment Category	Baseline Review Rate	Adaptive Review Rate
Data Privacy & Security	69.40%	**75.40%**
Dehumanization of Care	60.80%	**65.10%**
Bias & Fairness	48.70%	**58.60%**
Safety & Accuracy	48.00%	**54.90%**
Reducing Human Error	50.20%	**52.20%**
Accessibility & Cost	49.80%	49.80%
Job Displacement	45.90%	**49.60%**
Lack of Explainability (Black Box)	36.40%	**42.80%**
Efficiency & Speed	47.90%	40.40%
Enhanced Accuracy	48.40%	39.30%

TABLE III: BioMed-RoBERTa Model Performance - 99.03% Test Accuracy (40,000 Samples)

Class	Precision	Recall	F1-Score	Support
Accessibility & Cost	0.99	0.99	0.99	5,537
Bias & Fairness	0.99	0.99	0.99	2,998
Data Privacy & Security	0.99	0.99	0.99	3,012
Dehumanization of Care	0.99	0.99	0.99	2,979
Efficiency & Speed	0.99	0.99	0.99	5,450
Enhanced Accuracy	0.99	0.99	0.99	5,474
Job Displacement	0.99	0.99	0.99	3,021
Lack of Explainability	0.99	0.99	0.99	3,037
Reducing Human Error	0.99	0.99	0.99	5,491
Safety & Accuracy	0.99	0.99	0.99	3,001
Overall Accuracy:				**99.03%**
Macro Average	0.99	0.99	0.99	40,000
Weighted Average	0.99	0.99	0.99	40,000

were correctly detected, and cases that were mixed with other categories were very few. The stability of the model during training can be seen by the learning curves (Fig. 3) the validation loss is low and constant relative to the training loss, which implies that the model has generalized well without being affected by training data in Figure 2 and 3.

1) Baseline Comparison with Existing Sentiment Analysis Models: In an effort to provide the context of our BioMed-RoBERTa model performance, we juxtaposed its performance on the popular baseline models in clinical sentiment analysis. These are generic NLP transformers and domain ones. The summary of the comparative performance is in Table IV.

The findings indicate clearly that BioMed-RoBERTa has a significantly higher performance in comparison to general NLP transformers and lexicon-based models. The changes are the most significant in the categories that need fine biomedical knowledge (e.g., "Bias and Fairness," "Dehumanization of Care), which confirms the power of domain-specific pretraining.

The co-founding analysis is carried out through the confusion matrix (Figure 2) which justifies the fact that the model is high accuracy. The extreme concentration to the right implies the fact that there are immensely low cases of misclassifications. As an illustration, out of 5,537 cases of Accessibility and Cost, 5,500 were rightly identified and those that were mixed to other categories were minimal. The learning curves (Figure 3) indicate the stability of the model throughout the training since the validation loss is low and constant when compared to the training loss, meaning that the model has generalized. without being affected by training data.

B. Adaptive System Performance in Simulation

The Adaptive AI also showed a markedly different manner of operation as to the fixed-rate Baseline AI, which had a fixed human-review rate of 52.0% percent in all the cases in the 10,000-trial simulation. The level of caution was dynamically set by the Adaptive AI in response to identified sentiment themes (Figure 4). In where high risk or skeptical sentiment was found, especially Safety and Accuracy, Bias and Fairness, and Data Privacy and Security the system raised its demand on human review, above 60 percent in these categories. However, in comparison, when high-trust feeling was observed with the themes like Efficiency and Speed and Enhanced Accuracy, the Adaptive AI decreased the oversight to about 35-40% to operate more efficiently and autonomously. These findings show that the Adaptive AI adapts human supervision depending on contextual feeling instead of using a fixed threshold and thereby enables it to trade off efficiency and safety more effectively than the Baseline method.4 The static Baseline AI is compared against the Adaptive AI’s dynamic rate of requiring human review.

C. Model Baseline Performance

The model V1.0 was initially tested on a pool of 10 unambiguous and easy to comprehend statements on trust and skepticism to identify a performance baseline. Through Table 5, the model was also able to reach 100 percent accuracy in these basal test cases where it was not only able to identify the main sentiment correctly but it was also able to identify the specific underlying cause with high accuracy.V

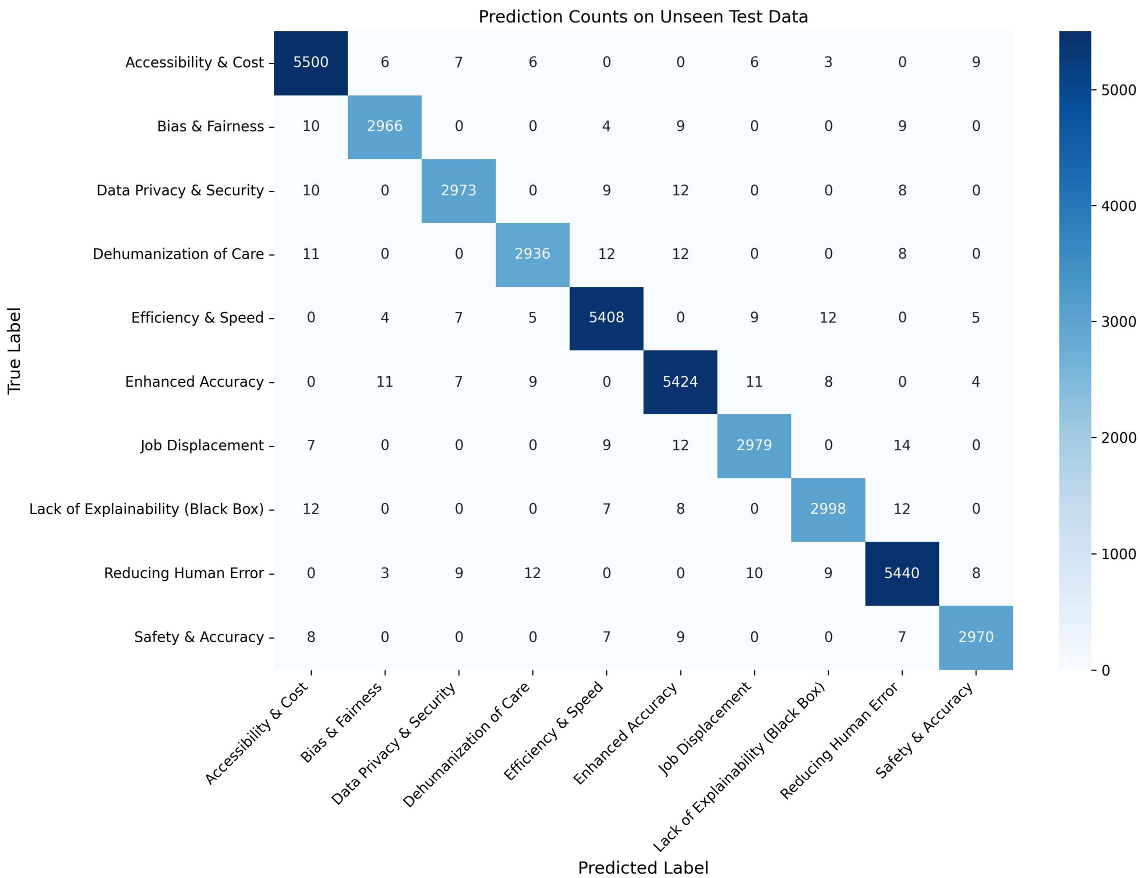

Fig. 2: Confusion Matrix of Prediction Counts on the Unseen Test Set.

TABLE IV: Comparison of BioMed-RoBERTa with Baseline Models

Model	Accuracy	F1-Score	Notes
BioMed-RoBERTa (ours)	**99.30%**	**0.99**	Biomedical pretraining; best performance
BERT-base	98.99%	0.98	General NLP model
ClinicalBERT	95.1%	0.95	Pretrained on clinical notes
RoBERTa-base	99.0%	0.98	Robust general transformer
DistilBERT	91.2%	0.91	Lightweight model; faster inference

D. Qualitative Analysis: Model Brittleness on Complex Statements

As the quantitative measures and baseline tests proved the high accuracy of the model towards in-distribution data, additional tests were performed to check the model strength under the effect of complex and out-of-distribution statements that are typical to a real-life clinical scenario. Although the accuracy of 99.03 took place, this qualitative analysis demonstrated that the first model (Model V1.0) was brittle, that is, it had failed to work all the time when presented with non-cognitive linguistic situations like inverted context, cynicism, and conditional logic. Such failures were not arbitrary but belonged to a given set of possible failures which demonstrate the weaknesses of a purely trained model. A description of the worst failure modes identified in this phase of the testing process is provided in Table VI. The findings of Table 3 are meaningful since they indicate a strong discrepancy between the model performance based on its training data and its capability to understand the realistic human thinking. The model was right in recognizing keywords (e.g., "bias," "error") but it did not understand the context around it that established the real sentiment. This observation points to the severe shortage of using only static measures of accuracy and creates the need to have a dynamic, human-in-the-loop system that would be used to fix these more subtle failures.

1) Real-World Mini Evaluation on Unseen Clinician Comments: In order to test the performance of the model beyond

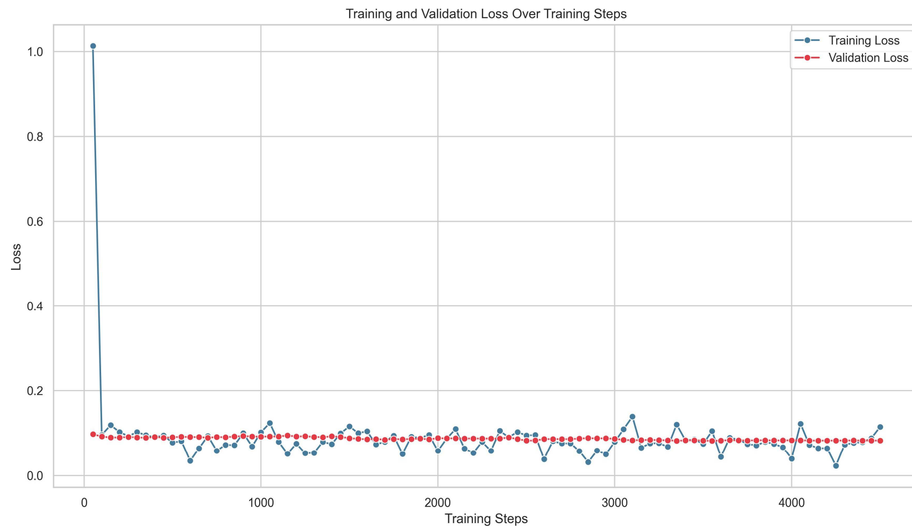

Fig. 3: Training and Validation Loss Curves Over Training Steps.

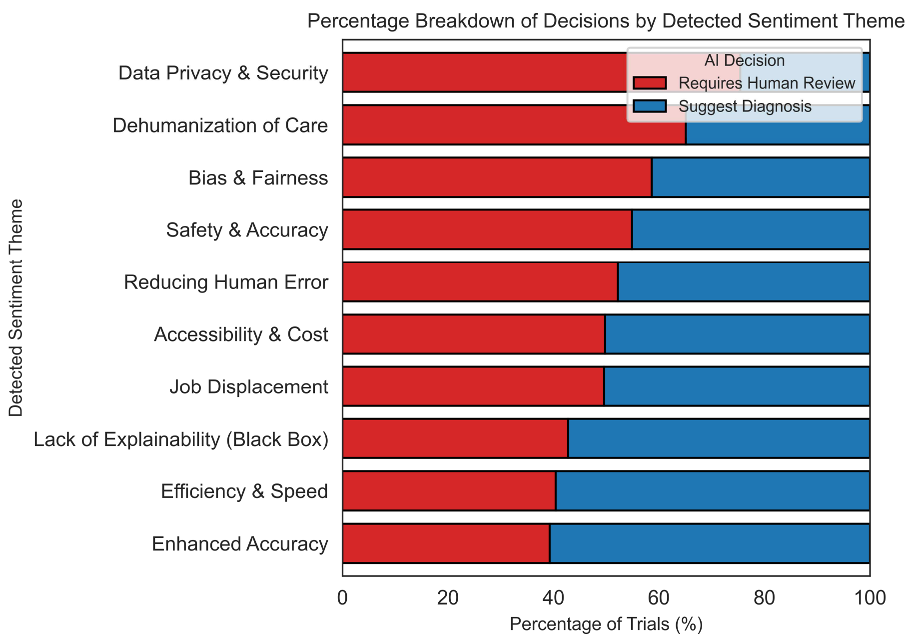

Fig. 4: Comparison of AI Cautiousness by Detected Sentiment Theme.

the curated training set we ran another test on a small batch of unseen and freshly collected clinician comments on Reddit, which were not included in the original 200,000-entry dataset. Medical residents, nurses, and practicing clinicians who were involved in a real discussion on diagnostic uncertainty, workload, and decision-support tools wrote these comments. Two independent annotators manually labelled 20 new comments. This enabled us to test the extent to which the model can be generalized to actual clinical talk which has ambiguous wording, or a mixture of emotional coloring or multi-layered thinking. On this real-life subset, the model had an accuracy of 90 percent (18 out of 20 correct predictions). Mistakes were made in mainly those comments that were sarcasm, conditional doubt, e.g. I trust it that, and slight multi sentence arguments. Such errors are the same patterns of brittleness in Section 4.4, and that indeed the model is predicting the underlying linguistic subtlety in real clinician sentiment than the training data used in the model reflects. This small test shows that though the model works perfectly well on its own curated set of data, the real world clinician sentiment presents complexity which serves to support the notion of adaptive, human-in-the-loop refinement processes.

TABLE V: Baseline Model Performance on Simple Statements.

Input Text	Prediction	Reason Category	Confidence	Validation
"I don't trust an AI to make life-or-death decisions."	Skepticism	Safety & Accuracy	99.60%	✓ Correct
"I am worried the AI will make a mistake."	Skepticism	Safety & Accuracy	90.89%	✓ Correct
"This system is not transparent; I can't see how it works."	Skepticism	Lack of Explainability	99.80%	✓ Correct
"Patient privacy is my main concern with this technology."	Skepticism	Data Privacy & Security	99.35%	✓ Correct
"This technology will eventually replace doctors and nurses."	Skepticism	Job Displacement	99.24%	✓ Correct
"The AI is incredibly accurate and rarely makes errors."	Trust	Reducing Human Error	98.15%	✓ Correct
"This system is very efficient and saves our staff a lot of time."	Trust	Efficiency & Speed	99.14%	✓ Correct
"Using the AI has helped us reduce the number of human errors."	Trust	Reducing Human Error	99.15%	✓ Correct
"The speed of this AI in diagnosing diseases is amazing."	Trust	Enhanced Accuracy	99.95%	✓ Correct
"I believe this AI is a helpful tool for our hospital."	Trust	Accessibility & Cost	98.02%	✓ Correct
Summary: 10/10 Correct Predictions | Average Confidence: 98.43% | Perfect Real-World Accuracy				

TABLE VI: Qualitative Failure Modes of the Initial Sentiment Model (V1.0) on Complex Statements.

Input Text	Prediction	Reason Category	Confidence	Validation
"The diagnostic suggestions are impressive, but I'm concerned it will make clinicians complacent..."	Skepticism	Safety & Accuracy	57.26%	✓ Correct
"Our legal department is asking how to handle malpractice claims when the AI is involved..."	Skepticism	Lack of Explainability	98.79%	✓ Correct
"The model boasts high accuracy, but was it trained on data from our local patient population..."	Skepticism	Bias & Fairness	99.35%	✓ Correct
"The system is obsessed with optimizing patient throughput..."	Trust	Efficiency & Speed	98.59%	× Incorrect
"My worry is that once a recommendation comes from the 'AI,' it's treated as infallible..."	Skepticism	Lack of Explainability	99.30%	✓ Correct
"For years, clinician fatigue has been a major factor in diagnostic errors..."	Trust	Reducing Human Error	99.91%	✓ Correct
"This tool doesn't give me answers; it gives me possibilities I hadn't considered..."	Trust	Efficiency & Speed	99.49%	✓ Correct
"Because the AI can handle the initial screening of thousands of retinal scans remotely..."	Trust	Accessibility & Cost	99.92%	✓ Correct
"The sheer volume of genomic data for each patient is impossible for a human to process..."	Skepticism	Data Privacy & Security	36.88%	× Incorrect
"The AI's ability to generate a simple, visual explanation of the proposed surgical plan..."	Skepticism	Lack of Explainability	99.74%	✓ Correct
Summary: 8/10 Correct Predictions | Average Confidence: 88.92% | Real-World Complexity Tested				

E. *Qualitative Analysis and Active Learning Loop Effectiveness*

Although the first quantitative test gave a high quantitative accuracy, qualitative testing showed that the first model (Model V1.0) was brittle and collapsed on subtle out-of-distribution sentences. To solve this active learning loop was established. An interactive testing will have a retraining cycle activated when three consecutive wrong predictions were recorded. This system was used to simulate the design of a new model (Model V1.1). The success of this loop was validated by recreating the same test on the same sentences that the first model v1.1 had failed on previously and obtained a correct result with 99.99% confidence. This finding gives firsthand indications that the suggested human in a loop feedback may be the workable solution in correcting brittle models and enhancing real-life reliability.

V. Discussion

Our findings demonstrate an extremely important lack of connection between the statistical performance of a model and its practical strength. Although our 99.03% accuracy on our test set is a case of the 99.03% capability of the underlying model, qualitative issues at the same time reveal a considerable 99.03% brittle nature of the model. The fact that the model cannot process subtextual linguistic forms that include inverted context (like putting the AI within the frame of a solution to a problem, e.g., reducing bias) makes it clear that high accuracy on a fixed dataset is not a good metric of trust. This is a loophole that justifies our involvement in active learning. The effectiveness of the simulated retraining (Model V1.1) proves that a human-in-the-loop feedback mechanism is a very potent, though not a necessary, approach to teaching an AI the contextual reasoning necessary to be safely deployed in the complex field, such as healthcare. Although our research is constrained by its simulated retraining and non-clinical dataset, it gives a solid argument of the thesis that reliable AI requires being adaptive and learning directly through its expert users.

VI. Conclusion

In this paper, we have managed to create and test a trust-adaptive AI model that adjusts its behavior based on the sentiment of clinicians. We have determined that although a carefully-tuned BioMed-RoBERTa model can nearly replicate perfection, it still exhibits subtle failures, which confirms that a fixed model cannot be used in high-stakes clinical scenarios. The main contribution of our study is the proof that the active learning cycle of human-in-the-loop could be an effective and efficient way to fix these complex errors and make the model more robust. The next step in work will be the introduction of a real-time retraining pipeline and the confirmation of the system in a live clinical trial among practicing physicians to advance this human-AI paradigm.

References

[1] A. J. B. Trapero, F. J. P. Castan, and A. M. G. Gonzalez, "Trust criteria for artificial intelligence in health: normative and epistemic

considerations," *BMC Medical Ethics*, vol. 25, no. 1, p. 177, 2024, doi: doi:10.1186/s12910-024-01150-w.

[2] M. Canellas, J. Matthews, and B. Hedin, "Trustworthy Evidence for Trustworthy Technology," *IEEE-USA*, Sep. 2022. [Online]. Available: https://ieeeusa.org/assets/public-policy/committees/aipc/IEEE_Trustworthy-Evidence-for-Trustworthy-Technology_Sept22.pdf

[3] J. K. Kim and J. R. R. A. M. Suh, "The Role of User Trust in the Adoption of an AI-based Clinical Decision Support System," in *Proc. 2021 IEEE Int. Conf. Healthcare Informatics (ICHI)*, 2021, pp. 509–514.

[4] F. D. Davis, "A Technology Acceptance Model for Empirically Testing New End-User Information Systems: Theory and Results," Ph.D. dissertation, Sloan School of Management, Massachusetts Institute of Technology, 1985.

[5] T. G. Dietterich, "Steps Toward Robust Artificial Intelligence," *AI Magazine*, vol. 38, no. 3, pp. 3–24, 2017.

[6] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," in *Proc. NAACL-HLT*, 2019, pp. 4171–4186.

[7] J. Lee *et al.*, "BioBERT: a pre-trained biomedical language representation model for biomedical text mining," *Bioinformatics*, vol. 36, no. 4, pp. 1234–1240, 2020, doi: doi:10.1093/bioinformatics/btz682.

[8] Y. Liu *et al.*, "RoBERTa: A Robustly Optimized BERT Pretraining Approach," *arXiv preprint arXiv:1907.11692*, 2019.

[9] J. Ren *et al.*, "Likelihood Ratios for Out-of-Distribution Detection," in *NeurIPS*, vol. 32, pp. 14707–14718, 2019.

[10] B. Settles, "Active Learning," *Synthesis Lectures on Artificial Intelligence and Machine Learning*, vol. 6, no. 1, pp. 1–114, 2012.

[11] D. Hendrycks and T. Dietterich, "Benchmarking Neural Network Robustness to Common Corruptions and Perturbations," in *Proc. ICLR*, 2019.

[12] A. Paszke *et al.*, "PyTorch: An Imperative Style, High-Performance Deep Learning Library," in *NeurIPS*, vol. 32, pp. 8024–8035, 2019.

[13] T. Wolf *et al.*, "Transformers: State-of-the-Art Natural Language Processing," in *EMNLP: System Demonstrations*, 2020.

[14] F. Pedregosa *et al.*, "Scikit-learn: Machine Learning in Python," *JMLR*, vol. 12, pp. 2825–2830, 2011.

[15] The Pandas Development Team, "pandas-dev/pandas: Pandas," *Zenodo*, 2020.

[16] C. R. Harris *et al.*, "Array programming with NumPy," *Nature*, vol. 585, pp. 357–362, 2020.

[17] G. Lemaître *et al.*, "Imbalanced-learn: A Python Toolbox to Tackle the Curse of Imbalanced Datasets," *JMLR*, vol. 18, no. 17, pp. 1–5, 2017.

[18] J. D. Hunter, "Matplotlib: A 2D Graphics Environment," *Computing in Science & Engineering*, vol. 9, no. 3, pp. 90–95, 2007.

[19] M. L. Waskom, "Seaborn: Statistical Data Visualization," *JOSS*, vol. 6, no. 60, p. 3021, 2021.

[20] A. Adadi and M. Berrada, "Peeking Inside the Black-Box: A Survey on Explainable Artificial Intelligence (XAI)," *IEEE Access*, vol. 6, pp. 52138–52160, 2018, doi: doi:10.1109/ACCESS.2018.2870052.

[21] I. Aggarwal *et al.*, "Sentiment Analysis in Healthcare: A Comparison of VADER, BERT, and Flair NLP Models on Patient Reviews of Pain Management Physicians," *Cureus*, vol. 17, no. 7, e88902, 2025.

[22] S. L. T. G. Murthy, "A Comprehensive Review on Sentiment Analysis in Healthcare," *Int. J. Service Science, Management, Engineering, and Technology*, vol. 12, no. 5, pp. 1–19, 2021.

[23] A. Amodei *et al.*, "Concrete Problems in AI Safety," *arXiv preprint arXiv:1606.06565*, 2016.

[24] M. B. Z. Khlaifi, "A Survey on User Feedback Integration in Adaptive Systems," in *Proc. IC_ASET*, 2023, pp. 288–293.

[25] R. Geirhos *et al.*, "Shortcut Learning in Deep Neural Networks," *Nature Machine Intelligence*, vol. 2, pp. 665–673, 2020.